EXPLICIT UPPER AND LOWER BOUNDS FOR DISTINCT CONSECUTIVE SUMS IN PERMUTATIONS

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Abstract. Let $f(n)$ be the largest number of distinct nonempty consecutive sums in a permutation of $\{1,\dots,n\}$. We prove $f(n) \le (4435834/10^7)n^2 + 1100n$ for every $n \ge 1$, and $f(n) \ge 0.309799n^2$ for every $n \ge 2^{4096}$. The upper bound uses coupled rearrangement inequalities and an exact integer certificate. The lower bound uses constant-sum triples ordered by affine images of primitive-root powers, with the zero field label retained, and Fourier completion with restricted translations. At length $3p$, for every prime $p \ge 2^{16}$, the construction gives $(803/288)p^2 - 50000p^{7/4}(\log p + 3)^8$ distinct sums; this expression becomes positive only at about $p = 2^{304.22}$, and gives the coefficient 0.309799 for $p \ge 2^{416}$. Dusart's explicit prime-interval theorem supplies the extension to all lengths. Both limiting coefficients improve those of Konieczny, the strongest coefficients we located in a bounded literature search completed on 12 October 2026. The search was not exhaustive and does not establish priority. Neither the limiting value of $f(n)/n^2$ nor its existence is determined.

1. Introduction and statements

For a permutation $a = (a_1, \dots, a_n)$ of $\{1, \dots, n\}$, put

$$\mathcal{S}(a) = \{a_{i+1} + \cdots + a_j : 0 \le i < j \le n\}, \qquad f(n) = \max_a |\mathcal{S}(a)|.$$

Intervals are linear and nonempty, and sums are counted by distinct values, without multiplicity. This extremal quantity occurs in Erdős Problem 34 [1]. All logarithms below are natural. Every error in our bounds is explicit.

Theorem 1 (Upper bound)**.** *For every integer $n \ge 1$,*

$$f(n) \le \frac{4435834}{10^7} n^2 + 1100n.$$

Consequently $f(n) \le 0.4436n^2$ for every integer $n \ge 66\,265\,061$.

Theorem 2 (Prime lengths)**.** *Let $p \ge 2^{16}$ be prime, $k = (p-1)/2$, $m = \lfloor p^{3/4} \rfloor$, and $I_p = \{k-m, \dots, k+m\}$. The explicit family $a(p,c,t)$ of Section 4, with $c \in \mathbb{F}_p^*$, $t \in I_p$, and the label $x_{p-1} = 0$ retained, satisfies*

$$\max_{c \in \mathbb{F}_p^*,\ t \in I_p} |\mathcal{S}(a(p,c,t))| \ge \frac{803}{288} p^2 - 50000p^{7/4}(\log p + 3)^8. \tag{1}$$

In fact the average over this parameter set has the same lower bound. For every prime $p \ge 2^{416}$, a member of this family has at least $0.309799(3p)^2$ distinct sums.

The threshold 2^{16} in Theorem 2 is a technical side condition, not an onset of numerical content. For an exact description, let $u_* > 29$ be the unique solution of

$$50000e^{-u_*/4}(u_* + 3)^8 = 803/288. \tag{2}$$

Within the theorem's range the right side of (1) is positive exactly when $p > e^{u_*}$. Numerically $u_*/\log 2 \approx 304.220$; a directed integer-series check encloses it strictly between 304.219 and 304.221. These numbers describe the real threshold, not a particular prime. Section 7 proves the exact criterion and the stated effective onsets. At all experimentally accessible lengths the error-bearing bound is vacuous.

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Theorem 3 (All lengths)**.** *For every integer* $n \geq 2^{4096}$,

$$f(n) \geq 0.309799n^2.$$

Consequently

$$0.309799 \leq \liminf_{n\to\infty} \frac{f(n)}{n^2} \leq \limsup_{n\to\infty} \frac{f(n)}{n^2} \leq 0.4435834.$$

The proof of Theorem 3 uses precisely the following published input, with the indicated endpoints:

(P) $$x \geq 3275 \quad \Longrightarrow \quad \text{there is a prime } p \text{ with } x < p \leq x\left(1 + \frac{1}{2(\log x)^2}\right).$$

This is Dusart's 1998 thesis, Théorème 1.9, printed p. 35 [6]. The prime-length result does not use this input. The maximizing parameters in Theorem 2 can be selected by finite search with an exact tie rule; no efficient algorithm at these large onsets is claimed.

2. Related work

Konieczny [2, Theorem 1.2, p. 415] proved

$$\frac{3}{2} - \frac{2}{\sqrt{e}} \leq \liminf_{n\to\infty} \frac{f(n)}{n^2} \leq \limsup_{n\to\infty} \frac{f(n)}{n^2} \leq \frac{1}{4} + \frac{\pi}{16}.$$

The endpoints are $0.286938680574\ldots$ and $0.446349540849\ldots$. The preprint first appeared in 2015; the journal version appeared in 2021. Our bounds improve the strongest coefficients we located in a bounded literature search completed on 12 October 2026, namely these coefficients of Konieczny. The search was not exhaustive and does not establish priority. It covered the problem's public records, relevant title and abstract searches, forward-citation records and selected public mathematical repositories. Such a search can miss unindexed work or work using different terminology.

Hegyvári's earlier quadratic-residue construction [3, Theorem 1, pp. 193–194] gives coefficient 1/18 after extension to a full permutation, as explained in [2, Section 1.5, p. 417]. Its short sequence has the form $a_{i+1} = 2p + r_{i+1} - r_i$, where $r_i = [i^2]_p$, and prefix sums $2pi + r_i$. Algebraic labels and distinct-difference methods are thus historical ingredients. The fixed-lag injectivity of primitive-root powers is also classical Welch–Costas machinery; see [4, Theorem 7.29] and the construction attributed there to Golomb [5]. We claim no originality for that ingredient.

Konieczny's lower construction randomizes complementary pairs [2, Section 4.2, Proposition 4.2]. The construction here uses complementary triples, a restricted translation average and explicit restoration of overlaps already covered by a base set. The upper reduction in [2, Section 5.1, (97)] counts low values and high-sum intervals. We retain this reduction and add coupled information about two high intervals and lower-threshold intervals. Sharpness of the auxiliary high-interval count is different from sharpness for distinct sums, as already emphasized in [2, Remark 5.2]. The original assertion $f(n) = o(n^2)$ was already disproved; the present paper sharpens coefficients and does not claim a solution of the remaining extremal questions.

3. The upper bound

3.1. **Rearrangement and the baseline deficit.** Fix a, use half-open position intervals, and let $s(X)$ denote the sum on a set of positions X. Set

$$T_n = n(n+1)/2, \quad U_n(r) = r(2n - r + 1)/2, \quad V(r) = r(r+1)/2.$$

Every r positions have sum in $[V(r), U_n(r)]$. Weighted rearrangement follows by swapping an inverted pair: if weights $c_i > c_j$ receive entries $a_i < a_j$, the swap gains $(c_i - c_j)(a_j - a_i) > 0$. This applies also to negative weights. Two length-k intervals at integer start separation $\delta \leq k$ therefore satisfy

$$s(\mathcal{A}) + s(\mathcal{B}) \leq U_n(k-\delta) + U_n(k+\delta) = 2nk - k^2 - \delta^2 + k. \tag{3}$$

Let H_k count length-k intervals with sum $> T_n/2$, and put

$$x_* = 1 - 1/\sqrt{2}, \qquad b(x) = \sqrt{1/2 - (1-x)^2}, \qquad g_0(x) = \begin{cases} 0, & x < x_*, \\ b(x), & x_* \leq x \leq 1/2, \\ 1-x, & 1/2 \leq x \leq 1. \end{cases}$$

Two such intervals cannot be disjoint. For $k \le n/2$, their extreme starts have separation δ satisfying

$$\delta^2 < n^2 b(k/n)^2 + k - n/2 \le n^2 b(k/n)^2.$$

Applying the same calculation to a single interval twice excludes $k/n < x_*$. A set of starts of span δ has at most $\delta + 1$ members, so

$$e_k = g_0(k/n) - (H_k - 1)_+/n \ge 0.$$

The function g_0 has variation 1 and integral $\pi/16$. Indeed, after putting $r^2 = 1/2$ and $u = 1 - x$, the triangle and circular segment have combined area $\pi r^2/8$; integration of the circular part uses $u = r\sin\theta$. For variation V_g, the normalized right-endpoint sum differs from the integral by at most V_g/n: on each cell bound the error by its width times its variation, and add. Consequently

$$\sum_k H_k \le (\pi/16)n^2 + 2n - n\sum_k e_k.$$

There are at most $T_n/2$ integer values at or below $T_n/2$; count higher values by representing intervals. With $B_0 = 1/4 + \pi/16$,

$$|\mathcal{S}(a)| \le B_0 n^2 + \frac{9}{4}n - n\sum_k e_k. \tag{4}$$

3.2. **Anchors and general complement inequalities.** Suppose a bin $[\alpha, \beta) \subset (x_*, 1/2)$ contains a length k/n with $e_k \le v < b(\alpha)$. Then $H_k \ge 2$. Choose its extreme high intervals $\mathcal{A} = [s, s+k)$ and $\mathcal{B} = [t, t+k)$, and set

$$x = k/n, \quad d = (t-s)/n, \quad I = \mathcal{A} \cap \mathcal{B}, \quad \mathcal{U} = \mathcal{A} \cup \mathcal{B}, \quad c = 1 - x - d.$$

Here $|I|/n = x - d$, $|\mathcal{U}^c|/n = c$, and

$$b(x) - v \le d < b(x), \qquad d \ge b(\alpha) - v > 0.$$

The positive rank deficit is

$$E_u = U_n(|I|) + U_n(|\mathcal{U}|) - T_n = n^2\{b(x)^2 - d^2\} + k - n/2 \le n^2\{b(x)^2 - d^2\}. \tag{5}$$

Also $s(I) > s(\mathcal{U}^c) \ge V(nc)$, since the anchor sums total $> T_n$.

Choose rational ε satisfying

$$0 < \varepsilon < 1/16, \qquad 4\varepsilon \le (1 - \beta - b(\beta))^2, \qquad \varepsilon \le 2(1/2 - \beta)^2. \tag{6}$$

Here $1 - \beta - b(\beta) > 0$: for $\beta < 1/2$, $(1-\beta)^2 > 1/4$ and $b(\beta)^2 = 1/2 - (1-\beta)^2 < (1-\beta)^2$. Put $Q = (1/4 - \varepsilon)n^2 + n/2$. Suppose two disjoint length-j intervals L, R have sums $> Q$. Let G be their intervening gap, $\mathcal{C} = (L \cup R)^c$, and

$$\ell = j/n = 1/2 - y, \qquad h = 2y, \qquad z = |G|/n.$$

Since $|\mathcal{C}| = nh$,

$$s(\mathcal{C}) < T_n - 2Q = 2\varepsilon n^2 - n/2, \qquad 0 \le y < \sqrt{\varepsilon}. \tag{7}$$

The latter follows from $s(\mathcal{C}) \ge n^2 h^2/2$. The function $1 - x - b(x)$ decreases, so (6) gives $h \le c$ and

$$s(I) > V(nc) \ge n^2 c^2/2 \ge 2\varepsilon n^2 > s(\mathcal{C}). \tag{8}$$

Thus G cannot contain I.

Use the integer counts $Z = |\mathcal{U}^c|$, $m_C = |\mathcal{C}|$, $J = |\mathcal{C} \cap \mathcal{U}|$, $K_C = |\mathcal{C} \cap I|$, and $W_a = 2nd$. Rearrangement with weights $\mathbf{1}_{\mathcal{A}} + \mathbf{1}_{\mathcal{B}} - \mathbf{1}_{\mathcal{C}}$ gives

$$\begin{aligned} &s(\mathcal{C}) > H_1 - E_u, \\ &H_1 = V(m_C - J) + Z(J - K_C) + V(J - K_C) + (Z + W_a)K_C + V(K_C). \end{aligned} \tag{9}$$

Indeed its weighted maximum is $U_n(|I| - K_C) + U_n(|\mathcal{U}| - (J - K_C)) - V(m_C - J)$. Subtracting from $U_n(|I|) + U_n(|\mathcal{U}|)$ proves the formula. Weights $\mathbf{1}_{\mathcal{A}} + \mathbf{1}_{\mathcal{B}} - 2\mathbf{1}_{\mathcal{C}}$ similarly give

$$\begin{aligned} &2s(\mathcal{C}) > H_2 - E_u, \\ &H_2 = V(m_C - J) + V(m_C - K_C) + V(Z + J) + V(Z + W_a + K_C) - V(Z) - V(Z + W_a). \end{aligned} \tag{10}$$

The weighted maximum here is $U_n(|I| - K_C) + U_n(|\mathcal{U}| - J) - V(m_C - K_C) - V(m_C - J)$. Listing the weights $2, 1, 0, -1, -2$ proves both identities, including the finite linear terms, without any assumption $j \ge k$.

Put $u = J/n$, $w = K_C/n$. Discarding nonnegative linear terms in (9) and using (5) yields

$$\frac{s(\mathcal{C})}{n^2} > \frac{(h-u)^2}{2} + c(u-w) + \frac{(u-w)^2}{2} + (c+2d)w + \frac{w^2}{2} - b(x)^2 + d^2. \tag{11}$$

For completeness, the corresponding expression from (10) is

$$\frac{2s(\mathcal{C})}{n^2} > \frac{(h-u)^2}{2} + \frac{(h-w)^2}{2} + cu + \frac{u^2}{2} + (c+2d)w + \frac{w^2}{2} - b(x)^2 + d^2. \tag{12}$$

This uses $V(Z+J) - V(Z) = ZJ + V(J)$ and its analogue with K_C. In (11), the derivatives before the constant term are $c-h+2u-w \geq 0$ and $2d-u+2w \geq 0$ on the feasible domain $w \leq u$, $w \geq \max(0, u-2d)$, since $h \leq c$. For the second derivative, when $u \leq 2d$ use $w \geq 0$; when $u > 2d$ use $w \geq u - 2d$. Lower bounds $u \geq z$, $w \geq (z-\eta)_+$, $0 \leq \eta \leq d$, can therefore be substituted together: minimize first over $w \geq \max((z-\eta)_+, u-2d, 0)$ and then over $u \geq z$. Along the moving edge $w = u - 2d$ the derivative is $c-h+2u \geq 0$; at $u = z$ the maximum is $(z-\eta)_+$ because $\eta \leq d \leq 2d$. The derivatives of (12) are $c-h+2u$ and $c+2d-h+2w$, also nonnegative. If $G \subset I$, substitute $u = w = z$ and use (7) to obtain

$$z^2 + (1-x+d-h)z + h^2/2 - b(x)^2 + d^2 < 2\varepsilon, \tag{13}$$

$$z^2 + (1-x-h)z + h^2/2 - \{b(x)^2 - d^2\}/2 < 2\varepsilon. \tag{14}$$

3.3. **Three bounds on a protruding wing.** If G extends to the left of I, then L is disjoint from $\mathcal{B}$; the right case is its reflection. Define

$$D_a = t - s, \quad R_0 = n - k - j, \quad V_* = (L \cup \mathcal{B})^c, \quad P = |V_* \cap (\mathcal{A} \setminus \mathcal{B})|.$$

Thus $0 \leq P \leq D_a$, $P \leq R_0$, and

$$s(V_*) < T_n/2 - Q = \varepsilon n^2 - n/4. \tag{15}$$

The weights $\mathbf{1}_{\mathcal{A}} + \mathbf{1}_{\mathcal{B}} - \mathbf{1}_{V_*}$ give

$$s(V_*) > P^2 + (|\mathcal{U}^c| - R_0)P + V(R_0) - E_u. \tag{16}$$

The expression before subtracting E_u is exactly $V(R_0 - P) + |\mathcal{U}^c|P + V(P)$. Two further rearrangements give

$$s(\mathcal{A}) + s(\mathcal{B}) + s(L) \leq U_n(k-P) + U_n(k+j+P), \tag{17}$$

$$s(\mathcal{A}) + s(\mathcal{B}) + 2s(L) \leq U_n(D_a - P) + U_n(k - D_a + j) + U_n(k+j+P). \tag{18}$$

The displayed arguments are the superlevel sizes of the weights. Their largest size is at most n because $P \leq R_0$. Normalize $\eta = P/n$, $r = 1-x-\ell$, and $U_0(v) = v - v^2/2$. The strict thresholds and (5) imply

$$\eta^2 + (\ell - d)\eta + r^2/2 - b(x)^2 + d^2 < \varepsilon, \tag{19}$$

$$\eta^2 + \ell\eta < U_0(x) + U_0(x+\ell) - 3/4 + \varepsilon, \tag{20}$$

$$\eta^2 + (x+\ell-d)\eta < U_0(d) + U_0(x-d+\ell) + U_0(x+\ell) - 1 + 2\varepsilon. \tag{21}$$

The omitted terms in (19) are the favorable $R_0/(2n^2) + 1/(4n)$ on the left. The right-hand corrections in (20) and (21) are respectively $(2x+\ell-2)/(2n) < 0$ and $(x+\ell-3/2)/n < 0$.

If $r^2/2 \geq \varepsilon$, (15) is impossible. Otherwise, when $P < D_a$, geometry gives

$$G \subset \mathcal{U}, \qquad |\mathcal{C} \cap \mathcal{U}| \geq |G|, \qquad |\mathcal{C} \cap I| \geq (|G| - P)_+. \tag{22}$$

To see this, let a_L be the end of L. A left protrusion has $a_L < t$; at least $\min(D_a, t - a_L)$ wing positions lie outside L. Thus $P < D_a$ forces $a_L > s$ and $t - a_L \leq P$. By (8) the other end of G is before the other end of I, proving (22). It also covers a gap wholly in the wing. Additional exterior pieces of $\mathcal{C}$ only increase the intersection counts. No such conclusion is used when $P = D_a$.

3.4. **The coupled retained-wing inequality.** Choose rational d_0 with $0 < d_0 \le b(\alpha) - v \le d$ and write

$$X = \beta, \qquad \mathfrak{b} = b(X)^2, \qquad D_0 = \mathfrak{b} - d_0^2, \qquad r_X = 1 - X - \ell.$$

Replacing x by X weakens the necessary inequalities. In (19), $r^2/2 - b(x)^2$ decreases with x; the right side of (20) increases with x; the derivative of left minus right in (21) is $\eta - 2(1-x-\ell) - d \le 0$, since $\eta \le d$. The derivative of the resulting gap lower bound is $-z - 2(1-x) < 0$. The analogous derivatives for (13)–(14) are $-z - 2(1-x)$ and $-z - (1-x)$.

If $r_X^2 \ge 2\varepsilon$, no protrusion on either side is possible; use (13)–(14) at $x = X, d = d_0$, as their left sides increase with d. Otherwise (6) implies

$$\ell > 1 - X - \sqrt{2\varepsilon} \ge X > b(X) > d. \tag{23}$$

Here $X^2 - b(X)^2 = 2(X - 1/2)^2 > 0$. In particular a protrusion forces $j > k$; shorter low intervals occur only in the complete-containment case, which needs no restriction on j.

For $A > 0$ put $\rho(A, N) = 0$ when $N \le 0$, and otherwise put $\rho(A, N) = (\sqrt{A^2 + 4N} - A)/2$. Define

$$\begin{aligned} t_S &= \rho(\ell - d_0, \varepsilon - r_X^2/2 + D_0), \\ t_1 &= \rho(\ell, U_0(X) + U_0(X + \ell) - 3/4 + \varepsilon), \\ t_2 &= \rho(X + \ell - d_0, U_0(d_0) + U_0(X - d_0 + \ell) + U_0(X + \ell) - 1 + 2\varepsilon). \end{aligned} \tag{24}$$

Let $t_* = \min(t_S, t_1, t_2)$.

Lemma 3.1 (Coupled retained wing)**.** *If $t_* < d_0$, then $q = d - \eta \ge d_0 - t_* > 0$.*

Proof. Inequality (20) gives $\eta < t_1$ (or is impossible if its right side is nonpositive). For (19), substitute $\eta = d - q$ to obtain the necessary inequality

$$F_S(d, q) = d^2 + \ell d - (d + \ell)q + q^2 + r_X^2/2 - \mathfrak{b} < \varepsilon.$$

On $q \le d_0$, $d \ge d_0$, this increases with d; at $d = d_0$ it decreases with q because $2q - d_0 - \ell < 0$ by (23). If $t_S < d_0$ and $q < d_0 - t_S$, it is at least ε, a contradiction. This also covers $t_S = 0$, since then $F_S(d_0, d_0) \ge \varepsilon$.

The identity $U_0(d) + U_0(X + \ell - d) = U_0(X + \ell) + (X + \ell)d - d^2$ turns (21) into

$$F_2(d, q) = d^2 - (X + \ell + d)q + q^2 < 2U_0(X + \ell) - 1 + 2\varepsilon.$$

Its derivative in d is $2d - q > 0$; at $d = d_0$ its derivative in q is $2q - X - \ell - d_0 < 0$. The same argument gives $q \ge d_0 - t_2$, including the zero-root case. Use whichever root attains t_*. This bounds the retained wing q, not η at a substituted diameter. □

When $q_0 = d_0 - t_* > 0$, the wing is nonsaturated and (22) applies. Substitution into (11) gives, for $z \ge \eta$, exactly $z^2 + (1 - x - h + q)z + h^2/2 - b(x)^2 + q^2$. For $z < \eta$, the true expression exceeds this by $(d + q)(\eta - z) \ge 0$. It increases in q, with derivative $z + 2q$. Thus in both cases

$$z^2 + (1 - X - h + q_0)z + h^2/2 - \mathfrak{b} + q_0^2 < 2\varepsilon. \tag{25}$$

If $G \subset I$, (13) at X and the actual d implies the same inequality since $q_0 \le d_0 \le d$ and its left side increases with that parameter. Right protrusions follow by reflection. If $t_* \ge d_0$, only the crude bound $z \le h$ is used.

3.5. **Counting at the lower threshold.** Let M_j count length-j intervals with sum $> Q$. Put $r_\varepsilon^2 = 1/2 + 2\varepsilon$ and

$$g_\varepsilon(x) = \begin{cases} 0, & x < 1 - r_\varepsilon, \\ \min\{1 - x, \sqrt{r_\varepsilon^2 - (1-x)^2}\}, & x \ge 1 - r_\varepsilon. \end{cases}$$

For overlapping or adjacent extreme intervals, (3) bounds their separation by $ng_\varepsilon(j/n)$; the remaining finite correction is $j - n \le 0$. Using one interval twice excludes $j/n < 1 - r_\varepsilon$. Otherwise (7) applies and the separation is $n(\ell + z)$. For $y = 1/2 - j/n \in [0, \sqrt{\varepsilon}]$,

$$M_j \le ng_\varepsilon(j/n) + 1 + nF(y), \qquad F(y) = (K_{\mathrm{span}}(y) - b_\varepsilon(y))_+, \tag{26}$$

where $b_\varepsilon(y) = \sqrt{1/4 + 2\varepsilon - y - y^2}$. An upper bound K_{span} is the minimum of applicable caps:

$$\begin{aligned} &1/2 + y; \\ &\ell + \rho(1 - X + d_0 - 2y, 2\varepsilon - 2y^2 + D_0), \quad \ell + \rho(1 - X - 2y, 2\varepsilon - 2y^2 + D_0/2) && (r_X^2 \ge 2\varepsilon); \\ &\ell + \rho(1 - X + d_0 - 2y - t_*, 2\varepsilon - 2y^2 + D_0 + 2d_0 t_* - t_*^2) && (r_X^2 < 2\varepsilon, t_* < d_0). \end{aligned}$$

Outside this y range no F is needed. If g_ε uses the total-start cap then $b_\varepsilon \ge 1/2 + y \ge K_{\mathrm{span}}$, so $F = 0$ and (26) remains valid. As before, g_ε has variation less than 2 and integral $\pi/16 + \pi\varepsilon/4$. Suppose an upper rectangle sum B_i for F uses grid $M_{\mathrm{grid}} = 100000$. Each grid cell contains at most $n/M_{\mathrm{grid}} + 1$ possible y values. Counting low values by Q and high values by intervals therefore gives

$$|\mathcal{S}(a)| \le \{B_0 - (1-\pi/4)\varepsilon + B_i\}n^2 + \{7/2 + M_{\mathrm{grid}}B_i\}n. \tag{27}$$

The linear terms are $n/2$ from Q, at most $2n$ from the Riemann error, n from the per-length allowance, and $nM_{\mathrm{grid}}B_i$ from grid occupancy.

3.6. **Certificate and exact verification.** The electronic certificate contains 892 disjoint bins. A row $[a_i, b_i, v_i', e_i']$ means

$$[\alpha_i, \beta_i) = [a_i/10000, b_i/10000), \quad v_i = v_i'/10^6, \quad \varepsilon_i = e_i'/4000.$$

They range from 2929/10000 to 3821/10000. The verifier checks their order and disjointness. We describe all directed-rounding conventions so that the finite certificate has a precise mathematical meaning.

At scale $R_* = 10^{12}$, round $b(\alpha_i)$ down and set the rational $d_0 = b(\alpha_i)_{\mathrm{down}} - v_i$. Round $b(\beta_i)$ up for (6), and round D_0 up. Subsequent formulas use this particular rational d_0; upward rounding of D_0 weakens the bounds. The positive exterior lower bound also has a direct rational check: if $x = R_*\beta_i$ and b_+ is the integer upward root, verify $R_* - x - b_+ > 0$ before squaring. The preserved certificate program does not assert that sign separately; the supplemental publication check does, and the positivity follows analytically above.

For a cell $[y_0, y_1] = [r/M_{\mathrm{grid}}, (r+1)/M_{\mathrm{grid}}]$, start with cap $1/2 + y_1$. If $(1/2 + y_0 - X)^2 \ge 2\varepsilon$, use the two complete-containment roots. Otherwise bound the three roots in (24) upward, using denominator coefficients at y_1 and numerators at y_0. The numerators decrease with y: r_X increases, while ℓ and the arguments of the increasing function U_0 decrease. The denominator coefficients decrease as well. If the minimum upward root $t_{\mathrm{up}} < d_0$, use the partial cap; otherwise keep the crude cap. A cell crossing the complete-containment boundary may still use the partial cap, since on its complete part the retained parameter is $d \ge d_0 \ge d_0 - t_{\mathrm{up}}$.

In the gap root use $h = 2y_1$ in the coefficient and $2\varepsilon - 2y_0^2$ in the right side. The term $2d_0t - t^2$ increases for $0 \le t < d_0$, so use t_{up}. Outside the root use $\ell \le 1/2 - y_0$. The preserved program further replaces the gap root by its maximum with t_{up}, a conservative extra allowance. Subtract $b_\varepsilon(y_1)$ rounded down and take the positive part, obtaining an upper bound U_{ir} on the cell. The final cell may extend past $\sqrt{\varepsilon}$; the bound is needed only on the intersection. Set $B_i = M_{\mathrm{grid}}^{-1}\sum_r U_{ir}$.

All substantive arithmetic is integer arithmetic. For example the scaled function is $U(v) = v - \lfloor v^2/(2R_*)\rfloor$, an upward bound; every square root is explicitly rounded up or down, every positive-denominator division is directed, negative root numerators are replaced by zero, and every root coefficient is asserted positive. Thus no floating-point value decides a certificate inequality. The resulting exact checks are

$$W_{\mathrm{def}} := \sum_i (\beta_i - \alpha_i)v_i = 0.002798309, \qquad \sum_i v_i = 27.98309, \tag{28}$$

$$B_0 - W_{\mathrm{def}} \le C_*, \qquad C_* = 4435834/10^7, \tag{29}$$

$$B_0 - (1-\pi/4)\varepsilon_i + B_i \le C_* \quad \text{for all } i, \tag{30}$$

$$9/4 + \sum_i v_i \le 1100, \qquad 7/2 + M_{\mathrm{grid}}B_i \le 1100 \quad \text{for all } i. \tag{31}$$

There are 17 430 410 rectangles: 8 938 271 partial, 1 221 306 crude-cap and 7 270 833 complete cells. For orientation, the largest B_i is approximately 0.01058603280678933, the largest coefficient in (30) is approximately 0.4435833245825063, and the deficit coefficient is approximately 0.44355123184936207. These decimals are descriptive; (29)–(31) use integer cross multiplication.

The verifier proves $3.141592653589793 < \pi < 3.141592653589794$ by Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, using 40 and 12 alternating-series terms and directed first-omitted-term bounds. Tangent addition gives $\tan(2\arctan(1/5)) = 5/12$, $\tan(4\arctan(1/5)) = 120/119$, and $\tan(4\arctan(1/5) - \arctan(1/239)) = 1$; the last angle is in $(0, \pi/2)$, proving the identity.

3.7. **Dichotomy and the linear term.** If no bin supplies $e_k \le v_i$, each bin contains at least $n(\beta_i - \alpha_i) - 1$ integer lengths. Disjointness and $e_k \ge 0$ give $\sum_k e_k \ge nW_{\mathrm{def}} - \sum_i v_i$, including bins whose lower count is negative. Equations (4), (29), and (31) prove $|\mathcal{S}(a)| \le C_*n^2 + 1100n$. Otherwise an anchor pair exists, and (27), (30), and (31) prove the same bound. This holds for every a and every positive integer n. Finally

$166 \cdot 66\,265\,061 = 11\,000\,000\,126 \geq 11\,000\,000\,000$, so $1100/n \leq 0.0000166$ for $n \geq 66\,265\,061$. This proves Theorem 1.

4. The lower construction and its base

Let $p = 2k+1$ be prime, $N = p-1$, $T = (9p+3)/2$, and g the least primitive root modulo p. For clarity, a finite subgroup of the multiplicative group of a field is cyclic: take, for each prime dividing its exponent, an element realizing the maximal prime-power part, and multiply the resulting coprime-order components. An element of order the exponent results. The polynomial root bound applied to $X^e - 1$, and Lagrange's theorem, force this order to be the group size. Thus g exists. Write $[x]_p$ for the least nonnegative residue and set

$$x_i = g^i \quad (0 \leq i < N), \qquad x_N = 0. \tag{32}$$

For $c \in \mathbb{F}_p^*$ and $t \in \mathbb{F}_p$ define

$$A_i = [cx_i + t]_p + 1, \quad C_i = [cx_i + t + k]_p + 1, \quad Y_i = p + C_i, \quad Z_i = T - A_i - Y_i. \tag{33}$$

The sequence $a(p,c,t)$ concatenates (A_i, Z_i, Y_i) for $i = 0, \ldots, p-1$, including the final zero label. Both A and C permute $1, \ldots, p$. If $A_i \leq k+1$, then $C_i = A_i + k$ and $Z_i = 6k + 5 - 2A_i$; otherwise $C_i = A_i - k - 1$ and $Z_i = 8k + 6 - 2A_i$. These ranges give respectively the odd and even integers in $2p+1, \ldots, 3p$. Thus $a(p,c,t)$ is a permutation of $1, \ldots, 3p$, and every triple sums to T.

Its prefix sums are

$$U_j = jT \ (0 \leq j \leq p), \quad V_i = iT + A_i, \quad W_i = (i+1)T - p - C_i \ (0 \leq i < p).$$

The positive chronological differences having an endpoint U form a base set $\mathcal{B}_0$ with

$$|\mathcal{B}_0| = 2p^2 + 3p. \tag{34}$$

Here is a direct count. Near hT, the central offsets are $+A_j$ for $j \geq h$, $0 \leq h < p$, and $-A_i$ for $i \leq p - h$, $1 \leq h \leq p$; indices are always in $[0, p-1]$. They give $p(p+1)$ values. In each outer interval $(hT+p, (h+1)T-p)$, $0 \leq h < p$, the values are

$$hT + p + C_i \quad (i \leq p - 1 - h), \qquad (h+1)T - p - C_j \quad (j \geq h). \tag{35}$$

These give another $p(p+1)$ values. Their two ranges are disjoint because $T > 4p$. The central bands $[hT - p, hT + p]$ and the outer intervals are disjoint. Finally the p positive multiples of T are distinct from all these values, proving (34).

Use the central candidates

$$dT + A_{i+d} - A_i, \qquad dT - C_{i+d} + C_i \quad (1 \leq d < N, 0 \leq i < N - d). \tag{36}$$

Each class is injective at fixed d, because its residue coefficient in g^i is $\pm c(g^d - 1) \neq 0$. Distinct central bands are disjoint. Use also the cross candidates

$$R_{d,i} = (d+1)T - p - C_{i+d} - A_i, \qquad Q_{d,i} = (d-1)T + p + C_i + A_{i+d}, \tag{37}$$

for $2 \leq d < N$, $d \neq k$, $0 \leq i < N - d$. The coefficients $\mp c(1 + g^d)$ are nonzero precisely when $d \neq k$, so each class is injective in its outer interval. For $2 \leq d < N$ there are $(N-1)(N-2)$ cross candidates; deleting $d = k$ removes exactly N. Their number is $p^2 - 6p + 7 \geq p^2 - 6p$. The omitted lag 1 costs only a linear term and makes later domains uniform. Central and cross regions are disjoint.

5. Fourier completion and the new central values

5.1. **Elementary Fourier norms.** On a finite abelian group use normalized Fourier coefficients and inversion $f(x) = \sum_\chi \widehat{f}(\chi)\chi(x)$, and define $\|f\|_A = \sum_\chi |\widehat{f}(\chi)|$. Convolution and the triangle inequality show that $\|fg\|_A \leq \|f\|_A \|g\|_A$. Affine pullback cannot increase this norm, since it only combines coefficients and attaches unit phases. Put $L = \log p + 3$.

On $\mathbb{Z}/q\mathbb{Z}$, $q \leq p$, any cyclic interval has norm at most L. Its zero coefficient is at most 1, and geometric summation with $\sin(\pi r/q) \geq 2r/q$, $0 \leq r \leq q/2$, bounds the coefficient at cyclic distance $r > 0$ by $1/(2r)$. Summation gives $1 + H_{\lfloor q/2 \rfloor} \leq 2 + \log q \leq L$. For representatives $x, y \in [0, q)$, both $\mathbf{1}_{x<y}$ and $\mathbf{1}_{x+y<q}$ have norm at most $2L$. Indeed, put $s(x) = [x]_q/q$. Then

$$\mathbf{1}_{x+y<q} = 1 - s(x) - s(y) + s(x+y), \qquad \mathbf{1}_{x<y} = s(y) + s(q-1-x) - s(q-1-x+y).$$

Every nonconstant coefficient of s has modulus $1/(q|1-e^{2\pi ir/q}|) \le 1/(4|r|_q)$, so its nonconstant mass is at most $H_{\lfloor q/2\rfloor}/2$. The displayed identities have constant term at most 1 and nonconstant mass at most $3H_{\lfloor q/2\rfloor}/2$, proving the bound.

An ordered triangle in an ordinary interval J has indicator $\mathbf{1}_J(u)\mathbf{1}_J(v)\mathbf{1}_{u<v}$ and hence norm at most $2L^3$. This remains true after independent translations of u, v. For $1 \le a, b \le p$, the triangle $a+b \le p$ is a reflected comparison, with norm at most $2L$. For $a+b \ge (3p+3)/2$, restrict $a \ge (p+3)/2$ and use $u = a-(p+3)/2$, $v = p-b$ and the comparison $v \le u$; the bound $2L^3$ is more than sufficient. We use the common bound $8L^3$ for all the two-variable indicators below.

5.2. **A graph-completion lemma.** Write $e_q(v) = \exp(2\pi iv/q)$. For an indicator R on $\mathbb{F}_p^2$, write $|R| = \sum_{X,Y} R(X,Y)$.

Lemma 5.1. *Let $\alpha, \beta \in \mathbb{F}_p^*$, $s \in \mathbb{F}_p$, and $F(X,Y,Z) = R(X,Y)\mathbf{1}_{Z=\alpha X+\beta Y+s}$, with $\|R\|_A \le K_R$. For $D \subset (\mathbb{Z}/N\mathbb{Z})^3$,*

$$\left| \frac{1}{N} \sum_{c\neq 0} \sum_{(d,i,j)\in D} F(cg^i, cg^{i+d}, cg^j) - \frac{|D|\,|R|}{p^3} \right| \le K_R(p^{3/2}+3p)\|\mathbf{1}_D\|_A. \tag{38}$$

Proof. The Fourier coefficient at (u,v,w) is $p^{-1}e_p(-ws)\widehat{R}(u+\alpha w, v+\beta w)$. Its total mass is $\|R\|_A$, and its mass on each of $u=0$, $v=0$, $w=0$ is $\|R\|_A/p$: the relevant maps to the two arguments of $\widehat{R}$ are bijections, using $\alpha\beta \neq 0$. Let $\chi_a(g^r) = e_N(ar)$ and $G(\chi,u) = \sum_{x\neq 0}\chi(x)e_p(ux)$. Completing an index character gives exactly

$$\begin{aligned} &\frac{1}{N}\sum_{c\neq 0}\sum_{d,i,j \bmod N} e_N(ad+bi+ej)e_p(c(ug^i+vg^{i+d}+wg^j)) \\ &\qquad = \mathbf{1}_{b+e=0}G(\chi_{b-a},u)G(\chi_a,v)G(\chi_e,w). \end{aligned} \tag{39}$$

Substitute $X = cg^i$, $Y = cg^{i+d}$, $Z = cg^j$; the residual character in c is $\chi_{-b-e}(c)$, proving the identity. At zero additive frequency a Gauss sum is N for the principal character and zero otherwise. The principal sum at nonzero frequency is -1. For nonprincipal χ, rescaling to $u=1$ gives

$$|G(\chi,1)|^2 = \sum_{z\neq 0}\chi(z)\sum_{y\neq 0}e_p((z-1)y) = N - \sum_{z\neq 0,1}\chi(z) = p.$$

Thus (39) is at most $p^{3/2}$ in modulus when u, v, w are nonzero. If exactly one is zero it is at most $Np \le p^2$. If exactly two are zero, the condition $b+e=0$ forces all three characters to be principal and the modulus is at most $N^2 \le p^2$. If all three are zero, only $a=b=e=0$ contributes, with value N^3. Fourier inversion gives the principal term $|D||R|/p^3$. The all-nonzero frequencies cost at most $K_Rp^{3/2}\|\mathbf{1}_D\|_A$; the other frequencies cost at most $3K_Rp\|\mathbf{1}_D\|_A$, using the three coordinate-plane masses. □

Every three-label index domain used below satisfies

$$\|\mathbf{1}_D\|_A \le 16L^5. \tag{40}$$

Here are explicit factorizations. For representatives modulo N put $U(a,b) = \mathbf{1}_{[a]_N+[b]_N<N}$ and $T_<(a,b) = \mathbf{1}_{[a]_N<[b]_N}$. Positive missing labels use $T_<(j,d)U(i,d)$; negative missing labels use

$$\mathbf{1}_{[3,N-1]}(d)U(i,d)T_<(N-1-j,d-2).$$

The latter says $j \ge N-d+2$ and has norm at most $4L^3$; the former has norm at most $4L^2$. The central-overlap domain has indicator

$$T_<(j,d)U(i,d)\mathbf{1}_{[0,k-1]}(d)\big(\mathbf{1}_{[0,k-1]}(i+d)+\mathbf{1}_{[k,N-1]}(i)\big). \tag{41}$$

The no-wrap factor makes the two summands disjoint. Its norm is at most $8L^4$; reflecting j preserves the bound. Cross-base domains have $d \in [2,N-1]$, $U(i,d)$, and one of $j \ge d$, $U(d-1,j)$, $U(d-2,j)$, $j \ge d-1$. Comparison complements have norm at most $1+2L \le 3L$, so these domains have norm at most $6L^3$. These prove (40). For $p \ge 2^{16}$, $p^{3/2}+3p \le 2p^{3/2}$. With $K_R \le 8L^3$, each such count has error at most

$$E_0 = 256p^{3/2}L^8. \tag{42}$$

5.3. **Separate central novelty, uniformly in translation.** At lag d, positive offset A_j is missing from $\mathcal{B}_0$ exactly when $j < d$; negative offset $-A_j$ is missing exactly when $j > p-d$. For the negative sign use only $j \in [N-d+2, N-1]$, an empty set when $d < 3$. Zero-label missing values are unused in this lower count. For either class in (36), equality to a signed missing label is a graph as in Lemma 5.1, with coefficients $\alpha, \beta \in \{1, -1\}$. Its two-variable indicator specifies the sign of a difference in $1, \dots, p$, has mass $p(p-1)/2$, and obeys the norm bound above, for every t. For example $A(Y) - A(X) = A(Z)$ gives $Z = Y - X - t - 1$ and $A(Y) > A(X)$. The other class uses $C(X) - C(Y)$.

The domain sizes are

$$M_+ = \sum_{d=1}^{N-1} (N-d)d = \frac{N(N^2-1)}{6}, \quad M_- = \sum_{d=3}^{N-1} (N-d)(d-2) = \frac{(N-2)((N-2)^2-1)}{6}.$$

Their sum is $p^3/3 - 2p^2 + 14p/3 - 4 \geq p^3/3 - 2p^2$. Thus each class has expected new count, averaging $c \neq 0$ with t fixed, at least

$$\frac{p-1}{2p^2}(p^3/3 - 2p^2) - 2E_0 \geq p^2/6 - 3p/2 - 2E_0.$$

There is a unique signed missing label for each value, and class injectivity prevents repetitions. Adding the two classes yields

$$\mathbb{E}_c(\text{separate central new counts}) \geq p^2/3 - 3p - 4E_0. \tag{43}$$

5.4. **Restricted translations and central overlap.** Set

$$m = \lfloor p^{3/4} \rfloor, \qquad I_p = \{k-m, \dots, k+m\}, \qquad M = 2m+1. \tag{44}$$

For $p \geq 2^{16}$ this is an ordinary interval in $[0, p-1]$ and $2m \leq k-1$. Indeed $p^{1/4} \geq 16$ implies $2p^{3/4} \leq p/8$, and $p/8 \leq (p-3)/2$ for $p \geq 4$. From now on averages are over $c \in \mathbb{F}_p^*$ and $t \in I_p$.

Equality between the central classes at the same lag forces the second start label to be $-g^i$. Thus $d < k$ and the possible starts are

$$i \in [0, k-d-1] \ \cup \ [k, N-d-1]. \tag{45}$$

Write $X = cg^i$, $Y = cg^{i+d}$, $t = k+e$, $u = [X+t]_p$, $v = [Y+t]_p$, and $b = [2t+k]_p$. Since $C(-X) = [b-u]_p + 1$, the lifts agree exactly when u, v lie in the same one of $[0, b]$ and $[b+1, p-1]$; the common lift is positive exactly when $u < v$. The condition $2|e| \leq k-1$ gives $b = k-1+2e$. The positive and negative common-lift indicators R_+, R_- on the whole field square have equal masses

$$\begin{aligned} |R_\pm| &= \binom{k+2e}{2} + \binom{p-k-2e}{2} \\ &= p^2/4 - p/2 + 4e^2 - 2e + 1/4 \leq p^2/4 + 4m^2 + 2m + 1. \end{aligned} \tag{46}$$

Each is the sum of two same-side ordered triangles; consequently $\|R_\pm\|_A \leq 4L^3$.

A positive common lift equal to a missing A_j gives $Z = Y - X - t - 1$; a negative one gives $Z = X - Y - t - 1$. For an upper bound enlarge the nonzero negative missing-label set to $j \in [N-d, N-1]$. Both signs now use (41), or its reflection, with domain size

$$\sum_{d=1}^{k-1} d(N-2d) = (k^3 - k)/3 \leq p^3/24.$$

The zero label can occur only for the negative missing sign and contributes at most p common values over all lags. Lemma 5.1 and (46) give

$$\begin{aligned} \mathbb{E}(\text{central overlap outside } \mathcal{B}_0) &\leq p^2/48 + (4m^2 + 2m + 1)/12 + p + 2E_0 \\ &\leq p^2/48 + p^{3/2} + p + 2E_0. \end{aligned} \tag{47}$$

Here $m \geq 1$ implies $4m^2 + 2m + 1 \leq 7m^2 \leq 12p^{3/2}$. Subtracting (47) from (43) proves

$$\mathbb{E}(\text{new central union}) \geq 5p^2/16 - 4p - p^{3/2} - 6E_0. \tag{48}$$

6. Cross terms and restoration

6.1. **Separate cross novelty.** Put $H = T - 2p = (5p+3)/2$. A collision of either cross class with the nearer base tail has form $a+b = C_j$, and one with the opposite tail has form $a+b+C_j = H$, where (a,b) is (A_i, C_{i+d}) or (C_i, A_{i+d}). Each congruence is a three-label graph with both coefficients nonzero. The ordinary triangle has mass $p(p-1)/2$. The opposite triangle $a+b \ge H-p$ has mass $k(k+1)/2 = (p^2-1)/8$: reflect from (p,p) to nonnegative coordinates whose sum is at most $k-1$. Both obey the established norm bound.

For R the nonzero ordinary base indices are $j \ge d$ and the opposite indices $j \le N-d$. For Q they are respectively $j \le N+1-d$ and $j \ge d-1$, always clipped to $[0, N-1]$. These are the four domain forms above. Include $d = k$ for upper bounds. For each of the ordinary and opposite totals, the two domain sizes sum to at most

$$\sum_{d=1}^{N-1} (N-d)(2(N-d)+2) = \frac{2(N-1)N(N+1)}{3} \le \frac{2}{3}p^3. \tag{49}$$

The zero label costs at most $4p$ collisions, by injectivity (one per lag, class and tail type). The ordinary loss is at most $p^2/3 + 2E_0$, and the opposite loss at most $p^2/12 + 2E_0$, by Lemma 5.1. Subtracting these and the zero-label allowance from at least $p^2 - 6p$ candidates proves, uniformly in t,

$$\mathbb{E}(\text{separate cross new counts}) \ge 7p^2/12 - 10p - 4E_0. \tag{50}$$

6.2. **Four and five labels under restricted translation.** Put $q = 4p$, $L_0 = \log p + 2$, and, for $r \in \mathbb{Z}/q\mathbb{Z}$,

$$K(r) = p^{-1}\sum_{x=1}^{p} e_q(rx), \quad B(r) = \min(1, 2/|r|_q)\ (r \ne 0), \quad B(0) = 1, \quad B_1 = \sum_r B(r), \quad B_2 = \sum_r B(r)^2.$$

Geometric summation gives $|K(r)| \le B(r)$, while

$$B_1 \le 1 + 4H_{2p} \le 5 + 4\log(2p) \le 4L_0, \qquad B_2 < 15. \tag{51}$$

The last inequality follows from $B_2 \le 1 + 8\sum_{j\ge1} j^{-2}$ and $\sum_{j\ge1} j^{-2} < 1 + 1/4 + \int_2^\infty x^{-2}\,dx = 7/4$. For representatives $1 \le x_j \le p$ define

$$F_4(x) = \mathbf{1}_{x_1+x_2+x_3+x_4=H}, \qquad F_5(x) = F_4(x_1,x_2,x_3,x_4)\mathbf{1}_{x_3+x_4=x_5}.$$

Congruences modulo q impose exactly these integer equalities, because every possible nonzero discrepancy has magnitude less than q. Expanding them gives

$$\widehat{F}_4(\xi) = q^{-1}\sum_a e_q(-aH)\prod_{j=1}^{4} K(a-4\xi_j), \tag{52}$$

$$\widehat{F}_5(\xi) = q^{-2}\sum_{a,b} e_q(-aH)K(a-4\xi_1)K(a-4\xi_2) \cdot K(a+b-4\xi_3)K(a+b-4\xi_4)K(-b-4\xi_5). \tag{53}$$

For every $s \in \mathbb{F}_p$,

$$\sum_{\sum\xi=s} |\widehat{F}_4(\xi)| \le B_1^4/p, \qquad \sum_{\sum\xi=s} |\widehat{F}_5(\xi)| \le B_1^5/q. \tag{54}$$

For (52) put $r_j = a - 4\xi_j$. A fixed tuple r obeys $\sum r_j = 4a - 4s \pmod q$ and allows at most four a; for each, the ξ_j are unique if they exist. For (53) use its five arguments r_j; then $\sum r_j = 4a + b - 4s$, so there are at most q choices of a, with b, ξ determined. This proves the bounds. On the exceptional subspaces, still for every plane $\sum\xi = s$,

$$\begin{aligned} \sum_{\sum\xi=s,\ \xi_5=0} |\widehat{F}_5| &\le 4B_1^5/q^2, \\ \sum_{\sum\xi=s,\ \xi_1=\xi_2=0} |\widehat{F}_5| &\le B_2B_1^3/q^2, \\ \sum_{\sum\xi=s,\ \xi_3=\xi_4=0} |\widehat{F}_5| &\le 3B_2B_1^3/q^2. \end{aligned} \tag{55}$$

In the first case $b=-r_5$ and $4a=\sum r_j+r_5+4s$, with at most four roots. In the second $r_1=r_2=a$ and $b=r_3+r_4+r_5-2a+4s$; summing the repeated factor gives B_2. In the third put $w=a+b=r_3=r_4$; then $3a=r_1+r_2+w+r_5+4s$. There is exactly one root since $\gcd(3,4p)=1$ for $p>3$. We retain the conservative factor 3 in (55).

The translation weight $w_I(t)=p/M$ on I_p, zero elsewhere, has mean one and

$$W_I:=\|w_I\|_A\le (p/M)L\le p^{1/4}L. \tag{56}$$

Indeed $2\lfloor p^{3/4}\rfloor+1\ge p^{3/4}$. Translation averaging attaches $\widehat{w}_I(-\sum\xi)$ to a Fourier character. Thus multiplying the plane estimates (54)–(55) by these absolute weights and summing over s costs at most the factor W_I. The zero frequency keeps weight one. This accounts for every nonzero total-frequency plane; no independence or fixed-translation equidistribution is assumed.

For an interval J of power indices put $S_J(v)=\sum_{j\in J}e_p(vg^j)$. Interval completion with the Gauss sums in (39), and additive orthogonality, gives

$$|S_J(v)|\le\sqrt{p}L_0\quad(v\ne0),\qquad \sum_{v\in\mathbb{F}_p}|S_J(v)|^2=p|J|. \tag{57}$$

For the first, the cyclic interval norm is at most $1+H_{N/2}\le L_0$ and completed Gauss sums have modulus at most $\sqrt{p}$. The second uses the distinctness of g^j.

6.3. **Total cross overlap.** Only R at lag d and Q at lag $d+1$ share an outer interval. Their equality is

$$A_i+C_{i+d}+C_u+A_{u+d+1}=H. \tag{58}$$

For an upper bound include every $0\le d\le p-3$, $0\le i\le p-2-d$, $0\le u\le p-3-d$. The tuple count satisfies

$$M_4\le\sum_{d=0}^{p-3}(p-1-d)(p-2-d)=p(p-1)(p-2)/3\le p^3/3.$$

After c-averaging the character at ξ, apart from a fixed unit phase, equals

$$\frac{p\mathbf{1}_{(\xi_1+g^d\xi_2)g^i+(\xi_3+g^{d+1}\xi_4)g^u=0}-1}{p-1}. \tag{59}$$

For $\xi\ne0$ at most $2p^2$ tuples satisfy the indicated equation. Unless both coefficients vanish, each i admits at most one u, or none if exactly one coefficient vanishes. Both coefficients can vanish at at most one lag unless $\xi=0$; that lag contributes at most p^2 tuples and all other lags at most p^2. The summed absolute average is thus at most $(2p^3+M_4)/(p-1)\le 4p^2$ for $p\ge3$. Equations (54), (56), and (51) bound the nonzero-frequency contribution by

$$4W_IpB_1^4\le 1024W_IpL_0^4. \tag{60}$$

Inclusion-exclusion for four positive summands bounded by p gives

$$|F_4|=\binom{5k+3}{3}-4\binom{3k+2}{3}+6\binom{k+1}{3}=\frac{23p^3+15p^2+p+9}{48}. \tag{61}$$

Three variables cannot exceed p simultaneously in this count. Multiplying $|F_4|/p^4$ by $M_4\le p^3/3$ gives at most $23p^2/144+p$. Hence

$$\mathbb{E}(\text{total cross intersection})\le 23p^2/144+p+1024W_IpL_0^4. \tag{62}$$

6.4. **A five-label estimate.** Let D be a subset of lag indices and let I_d,U_d,J_d be intervals of power indices, each of size at most N. Insert in F_5 the representatives in $1,\dots,p$ of

$$cg^i+t+\kappa_1,\quad cg^{i+d}+t+\kappa_2,\quad cg^u+t+\kappa_3,\quad cg^{u+d+1}+t+\kappa_4,\quad cg^j+t+\kappa_5,$$

where the shifts are fixed. Let $M_5=\sum_d|I_d||U_d||J_d|$. We claim

$$\left|\mathbb{E}\sum_{d,i,u,j}F_5-\frac{|F_5|}{p^5}M_5\right|\le 20000W_Ip^{3/2}L_0^6. \tag{63}$$

For a frequency ξ put $a_d = \xi_1 + g^d\xi_2$, $b_d = \xi_3 + g^{d+1}\xi_4$, $e = \xi_5$. If all three are nonzero, (57) and Cauchy–Schwarz give

$$\frac{1}{N}\sum_{c\neq 0}|S_{I_d}(ca_d)S_{U_d}(cb_d)S_{J_d}(ce)| \\ \leq \sqrt{p}L_0\frac{p\sqrt{|U_d||J_d|}}{N} \leq 2p^{3/2}L_0. \tag{64}$$

In fact the final factor 2 is unnecessary since $\sqrt{|U_d||J_d|} \leq N$; we retain it throughout. If exactly two coefficients are nonzero, the two-factor Cauchy–Schwarz bound with the remaining interval length is at most $2p^2$. If exactly one is nonzero, average before taking absolute values: each nonzero field power has character average $-1/N$, so the modulus is the product of the three lengths divided by N, at most $2p^2$.

Outside the union of the three subspaces in (55), both a_d, b_d are nonzero polynomials of degree at most one in g^d, and $e \neq 0$. At most two lags have a vanishing coefficient. Thus the summed character contribution is at most $2p^{5/2}L_0 + 4p^2$. On that union, for $\xi \neq 0$ all coefficients can vanish at at most one lag: two distinct such lags would force $\xi = 0$. That exceptional lag costs at most p^3, and all other lags at most $2p^2$ each, giving $3p^3$. Using the plane bounds, the nonzero-frequency error is at most W_I times

$$\tfrac{1}{2}p^{3/2}L_0B_1^5 + \tfrac{7}{4}pB_1^5 + \tfrac{3B_2}{4}pB_1^3 \\ \leq (512 + 1792 + 720)p^{3/2}L_0^6 < 20000p^{3/2}L_0^6.$$

The zero frequency supplies the stated main term, proving (63). Exchanging the first and second coordinate pairs merely exchanges the exceptional subspaces and obeys the same bound.

6.5. **Restoring intersections already in the base.** Count F_5 by $s = x_3 + x_4 = x_5$. Necessarily $k+2 \leq s \leq 2k+1$; there are $(s-1)(s-k-1)$ choices for the two pairs. Therefore

$$|F_5| = \sum_{s=k+2}^{2k+1}(s-1)(s-k-1) = \frac{k(k+1)(5k+1)}{6} = \frac{(p^2-1)(5p-3)}{48}. \tag{65}$$

Use the retained lags $d \in \{2,\dots,p-3\}\setminus\{k-1,k\}$, $I_d = [0, p-2-d]$, and $U_d = [0, p-3-d]$. At a cross overlap put $s = C_u + A_{u+d+1}$, $v = A_i + C_{i+d}$. Then $s+v = H$ and the common sum is $dT+p+s = (d+1)T-p-v$. It is already in $\mathcal{B}_0$ if either

$$s = C_j, \quad 0 \leq j \leq p-2-d, \qquad \text{or} \qquad v = C_j, \quad d \leq j \leq p-2. \tag{66}$$

These are valid subsets of (35), using only nonzero labels. The first branch deliberately drops the additional valid index $j = p-1-d$, to keep both branch lengths $|J_d| = p-1-d$. The branches are disjoint because $s, v \leq p$ together contradict $H > 2p$. The outer interval fixes d; injectivity fixes the R and Q indices and the covering base label. Thus each branch restores distinct values, without repetitions between branches.

Each branch has tuple count

$$M_5 = \sum_d (p-1-d)^2(p-2-d), \qquad p^4/4 - 6p^3 \leq M_5 \leq p^4. \tag{67}$$

For the lower bound begin with $\sum_{s=1}^{p} s^3 \geq \int_0^p x^3\,dx = p^4/4$. First use lags $1 \leq d \leq p-3$, corresponding to $2 \leq s = p-1-d \leq p-2$. Removing the top two cubes costs at most $2p^3$, and replacing cubes by $s^3 - s^2$ costs at most p^3 (the $s=1$ term becomes zero). Removing lags $k-1, k$ costs at most $2p^3$, and removing lag 1 at most p^3. This proves the lower bound; the upper follows from at most p terms, each at most p^3. Equation (65) gives $|F_5|/p^5 \geq 5/(48p^2) - 1/(6p^3)$. Using the lower bound on M_5 for its positive coefficient and the upper bound for its negative coefficient, each principal term is at least

$$\frac{5}{48p^2}(p^4/4 - 6p^3) - \frac{p^4}{6p^3} = 5p^2/192 - 19p/24.$$

Applying (63) twice and using $19p/12 \leq 2p$ proves

$$\mathbb{E}(\text{restored base intersections}) \geq 5p^2/96 - 2p - 40000W_Ip^{3/2}L_0^6. \tag{68}$$

The accounting identity behind this step is

$$|(R\cup Q)\setminus\mathcal{B}_0| = |R\setminus\mathcal{B}_0| + |Q\setminus\mathcal{B}_0| - |R\cap Q| + |R\cap Q\cap\mathcal{B}_0|.$$

The last term is bounded from below only by the two disjoint branches just proved; the subtraction uses the enlarged domain in (62).

7. Explicit errors, onsets and padding

7.1. **Combining the counts and selecting parameters.** The base, new central union and new cross union are disjoint. Add (34), (48) and (50), subtract (62), and restore (68). The quadratic coefficient is

$$2 + 5/16 + 7/12 - 23/144 + 5/96 = 803/288.$$

The error is at most

$$14p + p^{3/2} + 10E_0 + 1024W_I pL_0^4 + 40000W_I p^{3/2}L_0^6.$$

Using $E_0 = 256p^{3/2}L^8$, $W_I \le p^{1/4}L$, $L_0 \le L$ and $p, L \ge 1$, this is at most

$$(14 + 1 + 2560 + 1024 + 40000)p^{7/4}L^8 = 43599p^{7/4}L^8 \le 50000p^{7/4}L^8. \tag{69}$$

This proves the average assertion and hence (1). Define $a^{(3p)}$ by evaluating every $c = 1, \dots, p-1$ and every $t \in I_p$, taking a pair maximizing $|\mathcal{S}|$, and breaking ties lexicographically by (t, c). Compute m as the largest integer with $m^4 \le p^3$. All construction steps and comparisons can use integer arithmetic. This is an unambiguous finite definition.

To justify the exact positivity statement after Theorem 2, write $u = \log p$. The normalized error in units of p^2 is $50000e^{-u/4}(u+3)^8$. Its logarithmic derivative is $8/(u+3) - 1/4$, positive for $u < 29$ and negative for $u > 29$. At $u = 16\log 2$ its value is greater than $50000 \cdot 3^8/16 > 803/288$; also $16\log 2 < 29$. It tends to zero as $u \to \infty$. This proves existence, uniqueness and the exact criterion (2). The approximate exponent 304.220 is a numerical description only; it is not used to prove an onset. For a fully explicit vanishing-error estimate, when $u \ge 3$, $(u+3)^8 \le 2^8 u^8$ and $e^{u/8} \ge (u/8)^8/8!$, whence

$$50000e^{-u/4}(u+3)^8 \le 50000 \cdot 2^8 \cdot 8^8 \cdot 8!\, e^{-u/8}.$$

In particular no unspecified asymptotic error is needed for the leading coefficient 803/2592 at prime lengths.

7.2. **Certified prime-length onsets.** Put $x = \log_2 p$. Since $\log p \le \log_2 p$, the normalized error is at most $50000(x+3)^8 2^{-x/4}$. Its logarithmic derivative is $8/(x+3) - (\log 2)/4 < 0$ for $x \ge 416$, using $\log 2 > 2/3$. One elementary proof of the latter is the positive series $\log 2 = 2\sum_{j\ge 0}(1/3)^{2j+1}/(2j+1)$. The exact integer comparisons

$$50000 \cdot 419^8 \cdot 9000000 < 31 \cdot 2^{104}, \tag{70}$$

$$50000 \cdot 451^8 \cdot 90000000 < 67 \cdot 2^{112} \tag{71}$$

and the identities

$$\frac{803}{288} - 9\frac{309799}{10^6} = \frac{31}{9000000}, \qquad \frac{803}{288} - 9\frac{3097993}{10^7} = \frac{67}{90000000}$$

prove $|\mathcal{S}(a^{(3p)})| \ge 0.309799(3p)^2$ for $p \ge 2^{416}$, and the stronger $|\mathcal{S}(a^{(3p)})| \ge 0.3097993(3p)^2$ for $p \ge 2^{448}$. This proves Theorem 2. These onsets are sufficient and are not claimed minimal.

7.3. **Padding using the prime-interval theorem.** Let $n \ge 2^{4096}$, $b = \lfloor n/3 \rfloor$ and $x = (9999999/10000000)b$. Since $n \ge 30$, $b \ge 3n/10$ and $x > n/4 \ge 2^{4094} > 3275$. Thus $\log x > 4094(2/3) > 2700$ and $1/(2\log^2 x) < 1/14000000$. The rational inequality

$$\frac{9999999}{10000000}\frac{14000001}{14000000} < 1$$

shows that (P) supplies a prime in $(x, b]$. Choose the largest prime $p \le b$. Then $p > x > 2^{4094}$, in particular $p \ge 2^{448}$. Append $3p+1, \dots, n$ to $a^{(3p)}$; all old consecutive sums persist. Also $3b/n \ge 1 - 2/n \ge 999999999/1000000000$, so

$$\frac{3p}{n} > \frac{9999999}{10000000}\frac{999999999}{1000000000} > \frac{99999989}{100000000}.$$

Using (71), the padded permutation satisfies

$$\frac{|\mathcal{S}|}{n^2} \ge \frac{3097993}{10000000}\left(\frac{99999989}{100000000}\right)^2 > \frac{309799}{1000000}. \tag{72}$$

The last comparison is exact rational arithmetic. This proves the all-integer statement. Taking lower and upper limits and using $1100/n \to 0$ proves Theorem 3.

8. Computations and reproduction

The electronic bundle contains the complete upper table, the exact certificate program and its per-row output, the finite-check program and output, and all lower-construction programs and recorded outputs. The README gives exact commands and identifies every file; `SHA256SUMS` records their hashes. The supplied programs and outputs are preserved byte for byte. A separate small publication check verifies the positive exterior sign mentioned in Section 3.6 and the onset arithmetic. All programs use the Node.js standard library. The upper certificate uses arbitrary-precision integers, including directed square roots. Floating decimals in its output are summaries, not certificate inputs.

8.1. **Upper certificate and finite checks.** The 892-row, 17 430 410-cell certificate is part of the proof; the following finite permutation checks are supporting tests. All 46 233 permutations for $1 \le n \le 8$ were enumerated, giving

$$(f(1), \dots, f(8)) = (1, 3, 6, 9, 13, 19, 25, 32).$$

For every nontrivial extreme half-high pair with $k \le n/2$, the checker considers all disjoint equal-length low pairs, including lengths below k, for the general category inequalities and the three protrusion rearrangements when applicable. Additional tests use the tent $1, 3, 5, \dots$, followed by descending even entries, its reversal, and 40 seeded Fisher–Yates permutations at each $n \in \{16, 32, 64, 128, 256, 512\}$. The seed is hexadecimal 17d34210, and the threshold numerators are $80, 100, 120, 140, 160, 180, 200, 220$ over 4000. Three 3000-swap searches at each $n \in \{32, 64, 128\}$ retain counts $375, 1399, 5335$, with 9003 objective evaluations per length.

In total the preserved checker processes 46 488 permutations, 34 699 anchor pairs, 2 025 199 category tests, 2 312 221 tests of each protrusion rearrangement, 12 828 nonsaturated geometry cases, 10 999 saturated cases, 404 887 low rows and 458 coupled checks. The coupled trial loop imposes $j \ge k$; this is adequate for its protrusion branch because (23) forces $j > k$. The complete-containment proof itself imposes no such restriction. Finite tests cannot replace the universal rearrangement argument. For this bundle the exact certificate and this checker were replayed; their outputs agree with the preserved outputs after removing timing fields.

8.2. **Lower-construction checks.** Direct bitset enumeration covers all (c, t) for $p = 5, 7, 11, 17, 31, 61, 101$, including means and extrema by translation. A separate rotating-window multiset counter covers every (c, t) for $p = 5, 7, 11, 17, 31, 101, 251, 503, 1009$, a total of 1 343 802 pairs. Counts through $p = 31$ are cross-checked against the direct counter; the two independent scans also agree on the maximum and mean at $p = 101$. The maximum counts for $p = 251, 503, 1009$ are 176 678, 707 901, and 2 844 224 respectively. The largest exhaustively scanned prime is 1009.

The exact audit examines all 3190 pairs for $p = 5, 7, 11, 13, 17, 19, 23, 29, 31$. It checks the permutation, base cardinality, direct bitset count against a separate set enumeration, class injectivity, missing-label characterization, membership of candidates among interval sums, and 64 199 restored overlap values with disjoint restoration branches. It also checks 82 115 central-lift equivalences, 69 492 domain masks, 27 domain-size identities and 173 volume identities, together with (70)–(72) in arbitrary-precision integer arithmetic. This audit was replayed for the bundle and agrees with its preserved output apart from timing.

The recorded samples at $p = 251, 503, 1009, 2003, 3331$ use multipliers $1, 2, g, p - 1$ and translations $0, \lfloor p/8 \rfloor, \lfloor p/4 \rfloor, \lfloor 3p/8 \rfloor, (p-1)/2, \lfloor 5p/8 \rfloor, \lfloor 3p/4 \rfloor, \lfloor 7p/8 \rfloor$, for 160 counts. Padded examples give 30 981 553 sums at $n = 10000$, using $(p, c, t) = (3331, 3, 1665)$, and 69 813 779 at $n = 15010$, using $(5003, 2, 2501)$. These are counts for particular permutations, not the extremal values $f(n)$. These finite computations test combinatorial identities. They cannot test the content of (1) at its enormous onset, and finite unrestricted maxima above $0.31n^2$ do not prove an asymptotic gain. The theorem averages over the narrower set I_p.

The rotating counter is exact for the tested range. Multiplication $c = g^r$ rotates the N nonzero triple blocks, while the zero block stays last. It maintains the multiset of intervals inside that core, inside the zero block, and joining the core to the zero block. A one-block rotation removes the core intervals starting in the three departing positions and adds those ending in the three new positions, then replaces the joining intervals. Other sums persist. Changes of multiplicity across zero update the distinct-value count. The N rotations visit each nonzero multiplier once. This reduces the full affine scan from fourth-power enumeration to third-power multiset updates. The provided finite counters use fixed-width typed arrays and ordinary exact-range integer arithmetic; they are test programs for the documented sizes, not implementations for

the astronomical theorem onsets. The finite-search definition of $a^{(3p)}$ permits arbitrary-precision integer arithmetic instead.

9. Limits and unsuccessful routes

The bounds $f(n) \le 0.44n^2$ eventually and $f(n) \ge 0.31n^2$ eventually are not proved here. Nor do we prove that $f(n)/n^2$ converges. The following remarks describe obstructions encountered by these particular arguments, not impossibility theorems for stronger bounds.

Remark 9.1 (Upper-bound obstructions). The present table has $W_{\mathrm{def}} = 0.002798309$, whereas a deficit branch proving 0.44 would require $W_{\mathrm{def}} \ge B_0 - 0.44 > 0.006349540849362$. Its deficit exceeds 0.003551231849362, so these rows cannot prove that target. Averaging inequalities of the form $|\mathcal{S}|/n^2 \le B_0 - E$ and $|\mathcal{S}|/n^2 \le B_0 - A_x(e_x)$ adds no constraint: a saving s still requires the integrated inverse threshold $\int v_s(x)\,dx \ge s$, where $v_s(x) = \sup\{v : A_x(v) \ge s\}$.

High anchors alone are not mutually incompatible in the continuous relaxation. The symmetric tent $a(t) = 1 - 2|t - 1/2|$ has nested maximal cores and minimal exteriors. For symmetric windows of length x and span d, exterior mass is $((1-x)^2 + d^2)/2$ and common window mass is $1/2 - ((1-x)^2 + d^2)/2$; setting the latter to $1/4$ gives $d = b(x)$. Thus the high-interval envelope can be attained in that relaxation without information about low rows or distinct values.

Substituting an independent small diameter in the rank deficit and a large diameter in the wing is unjustified, as is replacing η by the third root evaluated at d_0. Lemma 3.1 instead bounds $d - \eta$. A saturated wing $P = D_a$ can allow the gap to extend outside the anchor union: (8) excludes containing the whole core but does not force a gap to meet the core. This is why nonsaturation is proved before using (22).

A floating scan using only the seed bound, at coefficient 0.4436 and bin width 0.0002, gave inverse-threshold integral about 0.00265469, below $B_0 - 0.4436 \approx 0.00274954$. This is exploratory evidence, not a certified limitation. Joint multipliers for two complements reproduced existing single-complement or mass-only constraints; no further retained-wing gain was established. A four-window floating relaxation used all nonzero weight vectors in $\{0,1,2,3,4\}^4$, normalized by their sum, gap increments 0.001, start increments 0.003, the parameter triples $(x, v, \varepsilon) = (.33, .03, .03), (.31, .07, .03), (.35, .015, .025), (.3, .09, .035)$, and $y \in \{.01, .03, .05, .07, .1\}$. Off-center configurations survived; for example the first triple at $y = .05$ allowed a tested gap .1 against a core-only root about .07998. Survival on a finite grid is not a feasibility proof and supplies no symmetrization lemma. Finally, an exploratory midpoint integration underestimated some rectangle bounds; those provisional rows were discarded or reduced, and only the passing directed-rounding certificate is used here.

Remark 9.2 (Lower-bound obstructions). The coefficient 803/2592 is below 31/100 by exactly 13/64800, or 13/7200 in units of p^2. Reducing the error term alone cannot close this quadratic gap. The minimum central lift mass in the present calculation is asymptotically $p^2/4$; no variance or positive-fraction argument supplying an additional fixed gain is proved here.

Fixing a translation in the four- and five-label estimates retains all total-frequency planes without the controlled weight (56). Deleting the translation average is therefore not justified. The restricted interval average, together with the estimates on every plane, is the proved replacement. Optimizing finite affine scans does not establish a uniform asymptotic gain, and unrestricted numerical maximizers are not substituted for the theorem's parameter set.

Four- and five-row alternatives were also tested. For odd $p = 2k+1$ and a permutation σ of $1, \dots, r$, take $z_1 = x$, $z_j = [2z_{j-1}]_p$ and the rows

$$(\sigma_1 - 1)p + x + 1, \qquad (\sigma_j - 1)p + k + 1 + z_j - z_{j-1}\ (2 \le j < r), \qquad \sigma_r p - z_{r-1}.$$

Since $[2z]_p - z$ permutes $[-k, k]$, these rows partition $1, \dots, rp$ and have constant total $r(rp+1)/2$. All row orders at $r = 4, 5$, $p = 101, 503$, $c = 1$, $t = 0, k$ give 576 recorded counts; 18 further counts use selected orders at $p = 1009, 3331, 5003$. The order $(2, 1, 3, 4)$ gives a finite ratio approximately 0.3125323102 at $p = 3331$ but 0.2895592821 at $p = 5003$. No uniform overlap estimate for the additional prefix classes was proved, so this route is neither established nor ruled out. Two interleaved triple families and a second-moment approach were considered, but no construction or estimate was completed for them.

AI use and verification status

The two proofs, the upper certificate and the accompanying programs were developed with GPT-6 Astra (OpenAI). Each proof was reviewed in isolation by two Claude (Anthropic) referees, both of whom returned PASS. A separate Claude verification process returned PASS for the upper certificate and PASS-WITH-REPAIRS for the lower result, with a wording-only correction concerning the lower inequality's vacuity. That correction is included in Theorem 2's discussion and Section 7. The expository corrections from the referees have also been incorporated. The review records report independent exact certificate checks and finite combinatorial tests; they do not constitute a formal proof verification. This manuscript was prepared with GPT-6 from the reviewed proofs. Its presentation has not undergone a separate referee round. No Lean formalization has been carried out. The bounded literature search described in Section 2 does not certify priority.

References

[1] T. F. Bloom, *Erdős Problems*, Problem 34, https://www.erdosproblems.com/34, accessed 12 October 2026 in the literature search described in Section 2.

[2] J. Konieczny, *On consecutive sums in permutations*, Journal of Combinatorics **12** (2021), no. 3, 413–477. Theorem 1.2, p. 415; Sections 1.5, 4.2 and 5.1. https://archive.intlpress.com/site/pub/files/_fulltext/journals/joc/2021/0012/0003/JOC-2021-0012-0003-a003.pdf. Preprint first posted 27 April 2015, https://arxiv.org/abs/1504.07156.

[3] N. Hegyvári, *On consecutive sums in sequences*, Acta Mathematica Hungarica **48** (1986), nos. 1–2, 193–200. https://doi.org/10.1007/BF01949064.

[4] K. Drakakis, *A review of Costas arrays*, Journal of Applied Mathematics **2006**, Article 26385, Theorem 7.29, pp. 20–21. https://www.pure.ed.ac.uk/ws/portalfiles/portal/10369967/A_review_of_Costas_arrays.pdf.

[5] S. W. Golomb, *Algebraic constructions for Costas arrays*, Journal of Combinatorial Theory, Series A **37** (1984), 13–21.

[6] P. Dusart, *Autour de la fonction qui compte le nombre de nombres premiers*, doctoral thesis, Université de Limoges, 26 May 1998. Théorème 1.9, printed p. 35 (supporting Proposition 1.10, p. 34). https://www.unilim.fr/pages_perso/pierre.dusart/Documents/T1998_01.pdf.

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